www.tntextbooks.in

etc. Respiration is aerobic through gills, skin, buccopharyngeal cavity and lungs. Vertebrates have a ventral muscular heart with two, three or four chambers and kidneys for excretion and osmoregulation.

Subphylum Vertebrata is divided into two divisions, **Agnatha** and **Gnathostomata**. Agnatha includes jawless fish-like aquatic vertebrates without paired appendages. Notochord persists in the adult. Gnathostomata includes jawed vertebrates with paired appendages. Notochord is replaced partly or wholly by the vertebral column. Agnatha includes one important class – Cyclostomata. Gnathostomata includes jawed fishes (Pisces) and Tetrapoda (amphibia, reptilia, aves and mammals). The superclass Pisces includes all fishes which are essentially aquatic forms with paired fins for swimming and gills for respiration. Pisces includes cartilaginous fishes (Chondrichthyes) and bony fishes (Osteicthyes).

2.4.4. Class: Cyclostomata

(G.*cyklos*–circle; *stomata* -mouth)

All members of cyclostomata are primitive, poikilothermic, jawless aquatic vertebrates and are ectoparasites on some fishes. Body is slender and eel-like bearing six to fifteen pair of gill slits for respiration. Mouth is circular without jaws and suctorial. Heart is two chambered and circulation is of closed type. No paired appendages. Cranium and vertebral column are cartilaginous. Cyclostomes are marine but migrate to fresh waters for spawning (anadromous migration). After spawning within a few days they die. The larvae (ammocoete) after metamorphosis returns to the ocean. Examples: *Petromyzon* (Lamprey) and *Myxine* (Hag fish) (Figure 2.23).

Lamprey

Hagfish

Figure 2.23 Examples of Cyclostomata

2.4.5. Class: Chondrichthyes

(*G. chondros* –cartilage; *ichthys* -fish)

They are marine fishes with cartilaginous endoskeleton. Notochord is persistent throughout life. Skin is tough covered by dermal **placoid scales** and the caudal fin is **heterocercal** (asymmetrical both externally and internally). Mouth is located ventrally and teeth are modified placoid scales which are backwardly directed. Their jaws are very powerful and are predaceous animals. Respiration by **lamelliform gills** without operculum (gill cover). Excretory organs are **mesonephric** kidneys. Two chambered heart is present. Cartilaginous fishes are ureotelic and store urea in their blood to maintain osmotic concentration of body fluids. They are **poikilothermic** and **viviparous**. Sexes are separate. In males pelvic fins bear claspers to aid in internal fertilisation.

Examples: *Scoliodon* (Shark), *Trygon* (Sting ray), *Pristis* (Saw fish) (Figure 2.24).